

The Unexplored Brain

New research is revealing that some well-known parts of the brain have unexpected functions. This is especially true of the “lower brain” areas, such as the brain stem and thalamus—areas once thought to be largely passive and to perform only automated roles.

Discovering potential

Cutting-edge scanning methods can probe areas of the brain beneath the cortex to understand their contributions to conscious thoughts and behaviors. These techniques include magnetoencephalography (MEG), which detects magnetic fields generated by neurons (see p.43), and fMRI and near-infrared spectroscopy (NIRS), which monitor brain activity by detecting changes in local blood flow and oxygenation.

The brain stem and emotion

Far from being a routine life-support region, the brain stem (see pp.36–37) is active in our behavior, especially emotions. Moods and feelings are even being localized to specific nuclei (clusters of nerve cells). These areas may be manipulated by electrodes or chemicals to treat problems such as depression, anxiety, and panic attacks.



Dorsal raphe

This nucleus is a major source of serotonin. Problems here can lead to worry, anxiety, and low mood.



Locus coeruleus

Malfunction of this major producer of noradrenaline may cause intense emotions, stress, and poor memory.



Pedunculopontine nucleus

This has roles in focused attention and concentration as well as in physical tasks, such as moving limbs.



Periaqueductal gray

Wrapped around the cerebral aqueduct channel, this nucleus is a major part of the pain-coping system.



Ventral tegmentum

This nucleus has a central function in motivation, learning, and reward and is implicated in conditions such as ADHD.

